Class 10 Math - Chapter 3: Pair of Linear Equations in Two Variables

🌟 Introduction

Two linear equations in the same two variables x and y are called a **pair of linear equations in two variables**. The general form of a pair of linear equations is:

1. $a_1x + b_1y + c_1 = 0$
2. $a_2x + b_2y + c_2 = 0$

Where $a_1, b_1, c_1, a_2, b_2, c_2$ are all real numbers, and $a_1^2 + b_1^2 \neq 0$, $a_2^2 + b_2^2 \neq 0$.

📌 Graphical Method of Solution

A pair of linear equations in two variables can perfectly map directly to two straight lines on a graph.

> [!NOTE]
>
> 1. **Intersecting Lines:** The lines intersect at exactly one point ($P(x,y)$). This represents a **unique solution**, and the equations are **consistent**.
> 2. **Parallel Lines:** The lines never meet. This represents **no solution**, and the equations are **inconsistent**.
> 3. **Coincident Lines:** The lines overlap exactly. This represents **infinitely many solutions**, and the equations are **consistent and dependent**.

Mind Map of Solutions

```
graph TD;
    Linear_Eq_Pair-->Intersecting_Lines;
    Linear_Eq_Pair-->Parallel_Lines;
    Linear_Eq_Pair-->Coincident_Lines;
    Intersecting_Lines-->Unique_Solution["Unique Solution\n(Consistent)"];
    Parallel_Lines-->No_Solution["No Solution\n(Inconsistent)"];
    Coincident_Lines-->Infinite_Solutions["Infinite Solutions\n(Dependent)"];
```

⚡ Algebraic Conditions for Consistency

You can determine the nature of the solution purely from the coefficients of the equations:

Ratio Comparison	Graphical Representation	Algebraic Interpretation
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Intersecting Lines	Unique Solution
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Parallel Lines	No Solution
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Coincident Lines	Infinitely Many Solutions

> [!IMPORTANT] The algebraic methods for solving pairs of linear equations are the **Substitution Method** and the **Elimination Method**. Make sure you are comfortable with both!

❌ Common Mistakes

> [!WARNING]

> 1. **Sign Errors:** When moving constant terms across the "$=$" sign, students frequently forget to change the sign. Always maintain algebraic balance. Example: $3x + 4y = 5 \Rightarrow 3x + 4y - 5 = 0$.
> 2. **Substitution Traps:** In the substitution method, when placing the value of x back into the second equation, ensure the **entire** polynomial is wrapped in parentheses before distributing. Example: substituting $x = 2y-1$ into $3x$ is $3(2y-1) = 6y - 3$, **not** $6y - 1$.
> 3. **Elimination Traps:** Remember that subtracting a negative number is equivalent to adding. Carefully watch your signs when subtracting columns during elimination!

📝 Quick Revision

The Substitution Method (Step-by-Step)

1. **Isolate:** Find the value of one variable (e.g., x) from one equation.
2. **Substitute:** Plug this value into the **other** equation to create an equation with only one variable.
3. **Solve:** Solve the single-variable equation.
4. **Plug Back:** Insert the found variable's value back into step 1's isolated equation to find the second variable.

Summary

Linear pairs govern real-world problems like calculating ages, determining train speeds, and combining mixtures. Whether graphical or algebraic, verifying the solution by plugging x and y back into **both** equations guarantees accuracy!